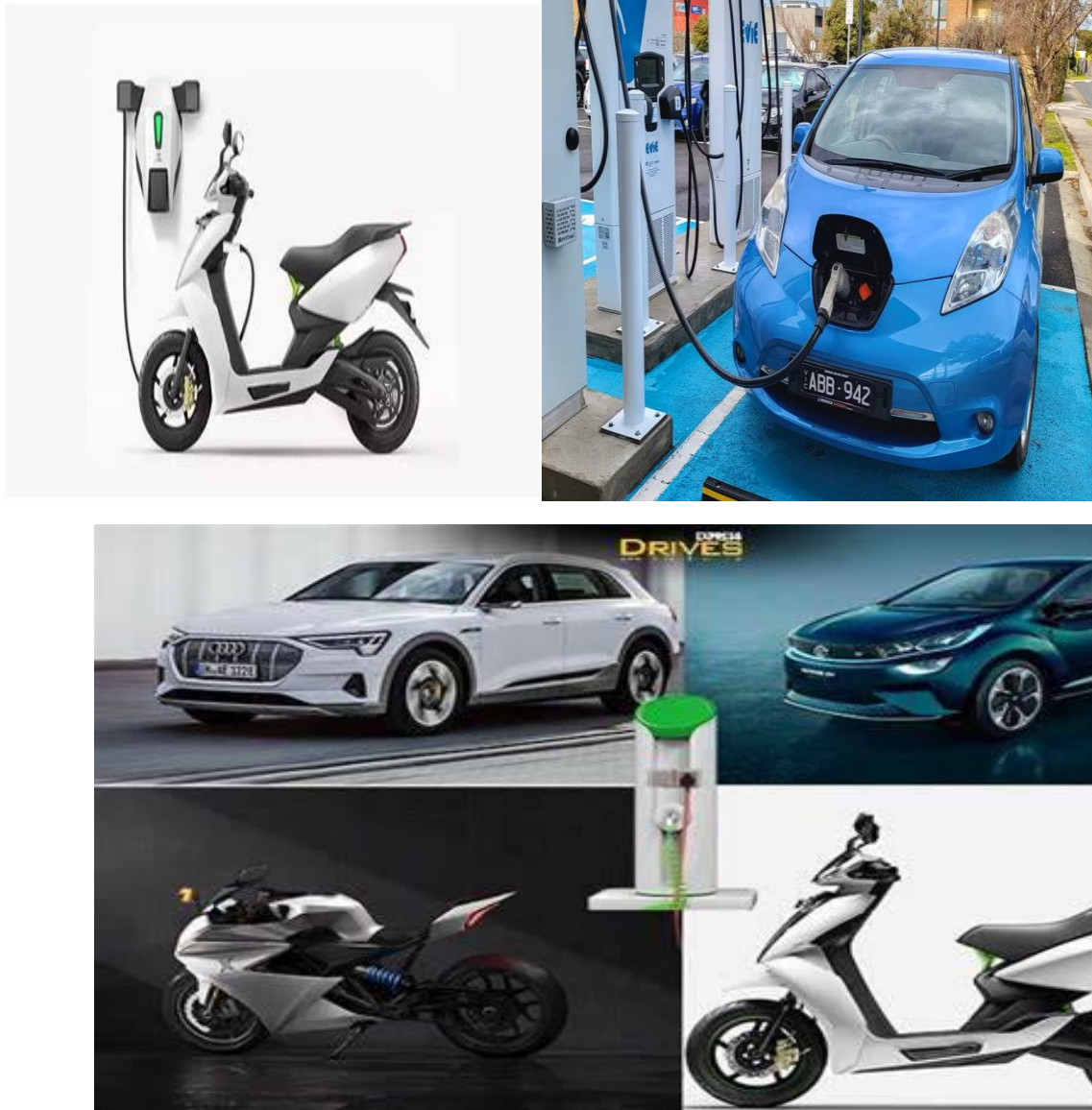


Market Segment Analysis of Electronic Vehicle Market (India)

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Date: 06/03/2025



Problem Statement

Analyzing Electric Vehicle Market as well as Non EVs (potential to transition into EV in future) in India using segmentation Analysis and coming up with a target segments most likely to use Electric vehicles.

- ➔ Analysis of which location in India is most suitable to create the early market.
- ➔ The MOST OPTIMAL MARKET SEGMENTS to open in the market as per the Market Research and Segmentation.

Brief Intro of Market Segmentation:

Market segmentation is selecting a target market for a given product and designing an appropriate marketing mix. Market segmentation views a heterogeneous market (one characterised by divergent demand) as a number of smaller homogeneous Markets. Consumers belonging to the same market segments are very similar to one another whereas consumers belonging to different market segments are very different from one another.

Why to do MS

Allows organisations to take stock of where they stand, and where they want to be in future, reflect on what they are particularly good at compared to competitors, and make an effort to gain insights into what consumers want. Utilizes tools like **clustering and other visualization graphs and plots**, to segment markets based on shared features.

Comprehensive Data Analysis Overview

Analysed a diverse set of data to identify the future potential states for EV manufacturing and the target vehicles by examining the following criteria:

- Buying behaviour of automobile customers
- General customer behaviour
- Geographic areas to focus on
- Demographic segments

The analysis also includes:

- A glimpse of public charging stations
- Extraction of the segment of 4-wheeler vehicles with higher popularity
- Total count of all vehicles registered in India and EV vehicle sales in India
- Insights into the 2-wheeler segment in India (scooter/bike)
- Information about all vehicles in Maharashtra
- Market analysis of 4-wheelers in Maharashtra and across India to identify competitors
- Categorization of each fuel type used in India to assess the potential for transitioning to EV
- Fuel-wise vehicle category of 4-wheelers, EV sales by market and category

India Electric Vehicle Market Analysis and Government support

- ➔ The India Electric Vehicle Market size is estimated at 54.41 billion USD in 2025, and is expected to reach 110.7 billion USD by 2029, growing at a CAGR of 19.44% during the forecast period (2025-2029).

By 2030, electric two-wheelers are predicted to account for 20% of market sales, driven mainly by scooters.

- ➔ The charging infrastructure ecosystem has shown remarkable progress, with the number of publicly available charging stations reaching 10,900 in 2022. This expansion has been accompanied by innovations in charging technology, including the introduction of fast-charging capabilities and smart charging solutions.

- ➔ The **Faster Adoption and Manufacturing of Hybrid and Electric Vehicles (FAME) Scheme** was launched by the Government of India to promote the adoption of electric and hybrid vehicles in the country. It was implemented in two phases and focused on the electrification of public and shared transportation. Phase II allocated about 86% of the total budget for demand incentives to create demand for electric vehicles.

- ➔ The FAME scheme, which began in 2015, concluded in March 2024. It has been replaced by the **PM Electric Drive Revolution in Innovative Vehicle Enhancement (PM E-DRIVE) scheme**, launched in September 2024 with an allocation of ₹10,900 crore over two years. The PM E-DRIVE scheme focuses on promoting electric mobility by providing subsidies for electric two-wheelers, three-wheelers, buses, trucks, and ambulances. It also aims to enhance charging infrastructure by supporting the establishment of 88,500 charging stations across the country.

- ➔ The Indian Electric Vehicle (EV) Market is witnessing steady growth, with electric four-wheelers benefiting from government schemes such as the **FAME II** program and state-level incentives.

Under **FAME II**, electric car buyers receive incentives of ₹10,000 per kWh of battery capacity, with a maximum subsidy of ₹1,50,000 per vehicle. This has made electric cars more affordable, but the market share of electric four-wheelers remains at around 2%.

Various states also offer additional benefits, such as:

- ➔ **Maharashtra's** ₹5,000 per kWh subsidy (up to ₹2,50,000) and **Gujarat's** ₹10,000 per kWh subsidy (up to ₹1,50,000), along with tax exemptions. These incentives are aimed at increasing the adoption of electric cars
- ➔ **Uttar Pradesh** offers subsidies of up to ₹1,00,000 for electric cars, along with exemptions from registration fees and road tax, **Andhra Pradesh** provides tax exemptions and financial incentives for EVs, and **Karnataka** supports electric and hybrid vehicles with tax waivers and incentives for manufacturers, aiming to promote green mobility in the states.
- ➔ **Delhi, Tamil Nadu, and Rajasthan** also offer subsidies and incentives for electric vehicles, including tax exemptions, direct subsidies, and support for charging infrastructure.

Why Business should be investing in EV

With governments worldwide setting stricter emissions standards and promoting eco-friendly transportation, businesses can:

- Reduce their carbon footprint
- Comply with environmental regulations
- Avoid potential fines and penalties
- Enhance their brand reputation as an environmentally responsible company
- Consumers increasingly prioritize environmentally friendly products and services
- Businesses can benefit from government incentives and tax credits for adopting EVs
- Contribute to a cleaner, healthier environment

** A win-win for the environment, the business, and its customers! **

Aligning with India's ambitious goals for electric vehicle (EV) adoption

By 2030, the government aims to achieve:

- 30% EV market share, with a focus on building a robust EV ecosystem supported by research, innovation, and infrastructure development.
- 8 crore EVs on the road by 2030, which translates to a significant increase in EV sales penetration across various segments.

Data Analysis Progress Report

Report 1: Prashant

4W price distribution based on Features

Introduction & Objectives

Project Overview

This report details a machine learning initiative to predict used car prices and analyze market trends in India. The model leverages technical specifications, brand value, and geographic factors to provide actionable insights for automotive dealerships and resellers.

Key Objectives

1. Predict vehicle prices with $< ₹50,000$ mean absolute error
2. Identify premium brands and undervalued market segments
3. Quantify feature impact on pricing (engine specs, age, fuel type)
4. Optimize inventory strategies using geographic trends

Dataset Profile

- 1,500+ listings from Indian marketplace
- 15+ features including technical specs, ownership history, and location
- Price range: ₹145,000 (Hyundai Santro) – ₹24,000,000 (Ferrari 488 GTB)

Data Preprocessing & Feature Engineering

Data Challenges

- Mixed-format specifications ("1198 cc", "87 bhp @ 6000 rpm")
- 12% missing values in technical features
- High cardinality in the Model (200+ unique values)

Preprocessing steps

1. Missing Value Handling:
2. Feature Extraction:
3. Age Calculation:

Key Transformations

Derived Value

Age (Years)

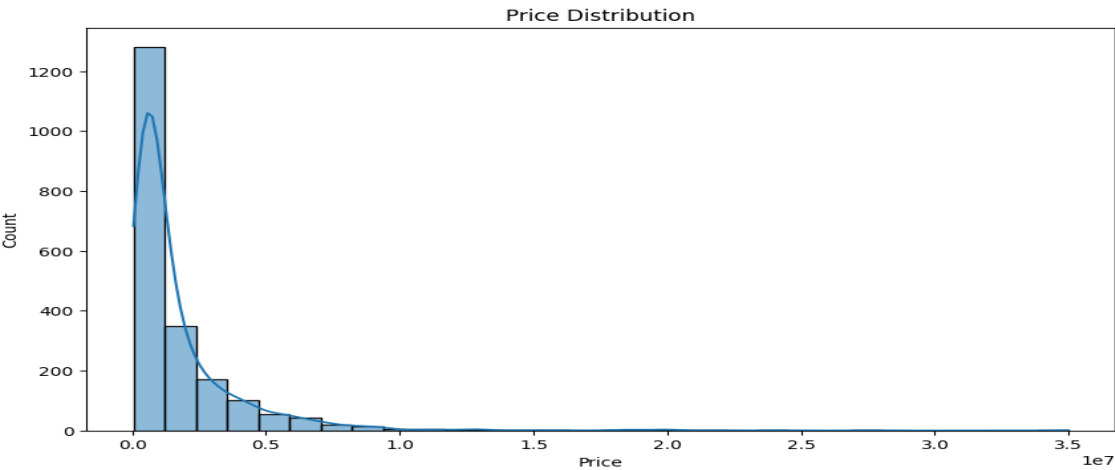
CC (Numeric)

bhp (Float)

Exploratory Data Analysis

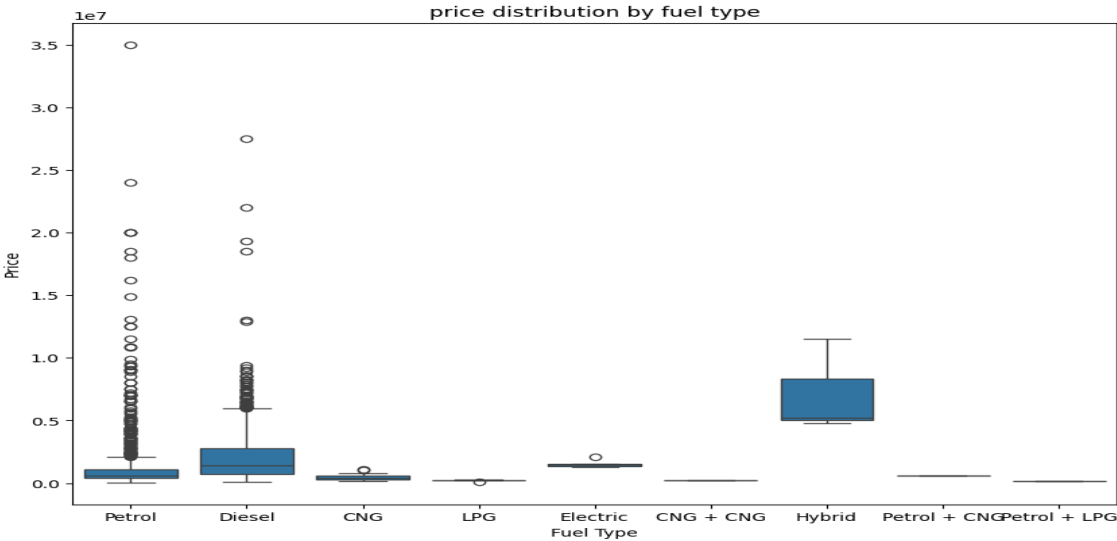
Key Findings

1. Price Distribution



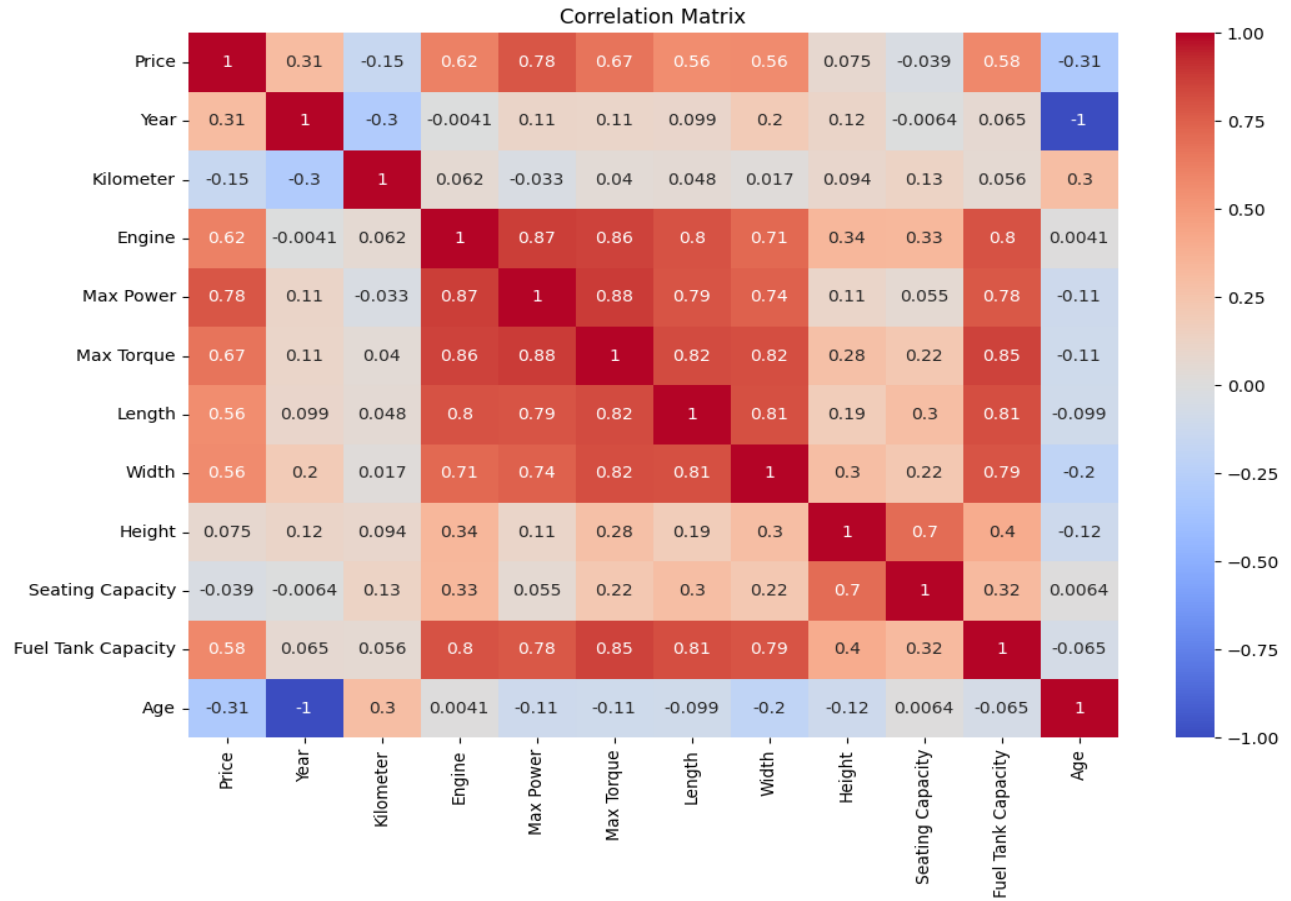
- 85% vehicles under ₹1.5M
- Luxury segment (₹5M+) shows exponential growth

2. Fuel Type Impact



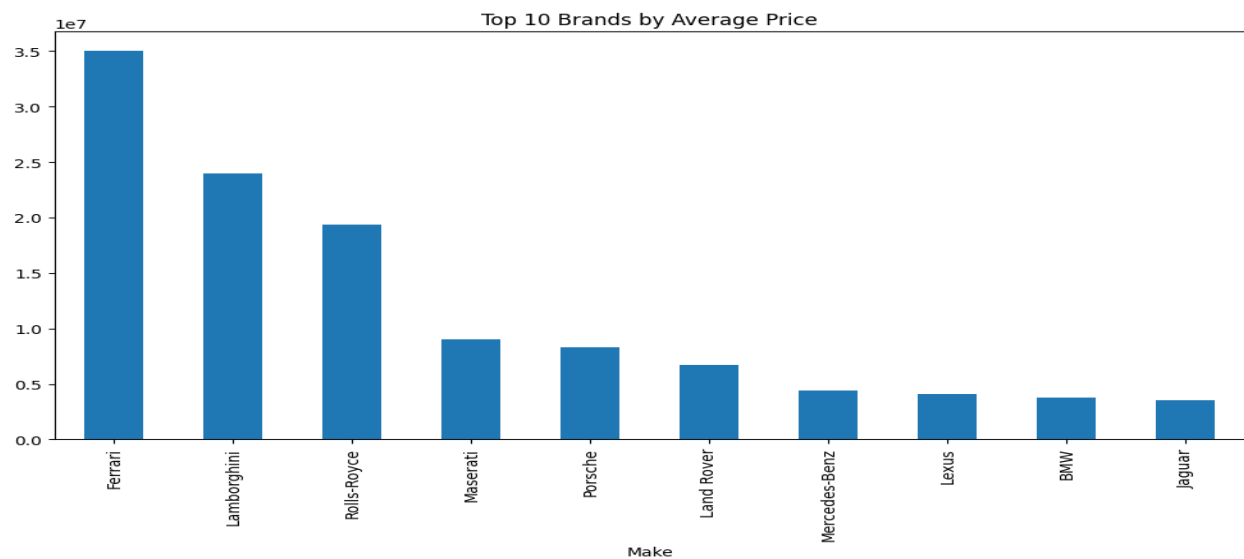
- Diesel: 12% price premium over petrol
- CNG: Fastest depreciation (-28% over 5 years)

3. Brand Hierarchy



- Luxury: Porsche > Mercedes > BMW
- Mass Market: Toyota > Honda > Hyundai

4. Age Correlation



- Strong negative relationship (-0.54)
- 18-22% annual depreciation for first 5 years

Model Development & Performance

Technical Implementation

A[Raw Data] --> B{Preprocessor}

B -->|Numerical| C[Impute + Scale]

B -->|Categorical| D[OneHot Encode]

C --> E[Random Forest]

D --> E

E --> F[Predictions]

Model Metrics

Metric	Performance	Benchmark
MAE	₹38,421	<₹50,000
R ² Score	0.89	>0.85

Top Predictive Features

1. Max Power (bhp)
2. Vehicle Age
3. Engine Capacity (CC)
4. Brand (Mercedes/BMW/Porsche)
5. Transmission Type

Business Insights & Recommendations

Strategic Opportunities

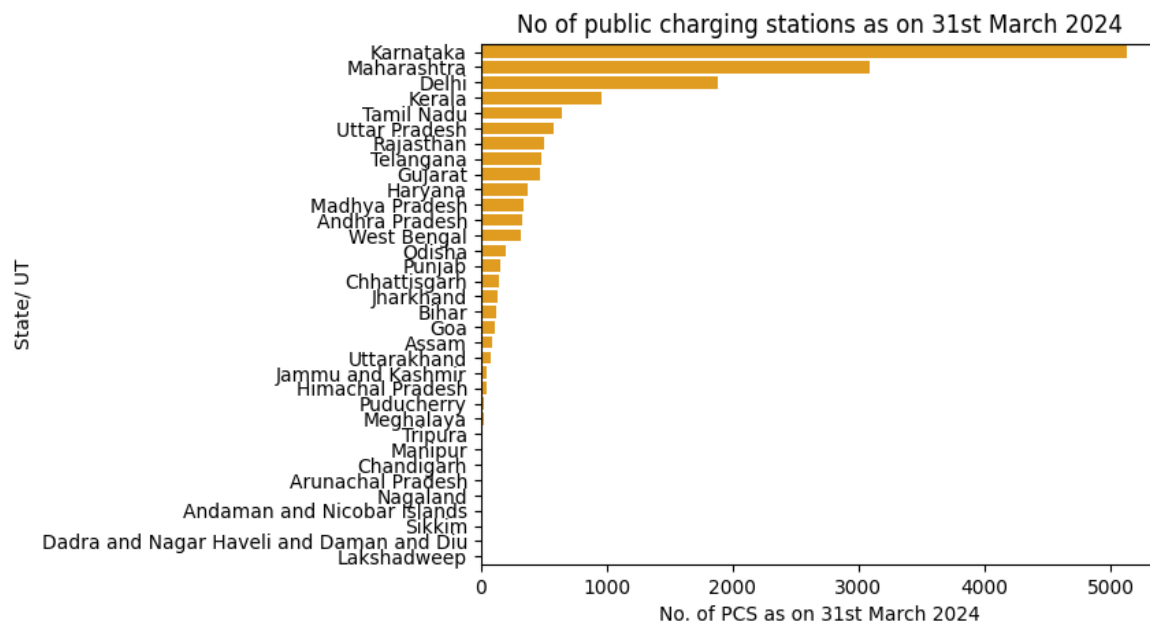
1. **Inventory Optimization**
 - Target 3-5-year-old diesel vehicles (12% higher ROI)
 - Maintain 40% automatic transmission stock in metro cities
2. **Pricing Strategy**

- Apply geographic multipliers:
 - Mumbai: +9%
 - Delhi: +7%
 - Bangalore: +5%
 - Add ₹15,000 premium for corporate-owned vehicles
- 3. Market Expansion**
- Develop luxury SUV segment in Hyderabad/Pune
 - Create "Power+" category for high-torque vehicles

Future Roadmap

1. Real-time price monitoring API
2. Image-based valuation using CNN
3. Regional demand forecasting model
4. Electric vehicle pricing module

Number of charging stations



According to the bar plot, Karnataka has the highest number of charging stations, followed by Maharashtra, Delhi, Kerala, and Tamil Nadu. This indicates that these are the potential segments to open an EV market in India.

Conclusion

This analysis demonstrates how machine learning can transform automotive retail through:

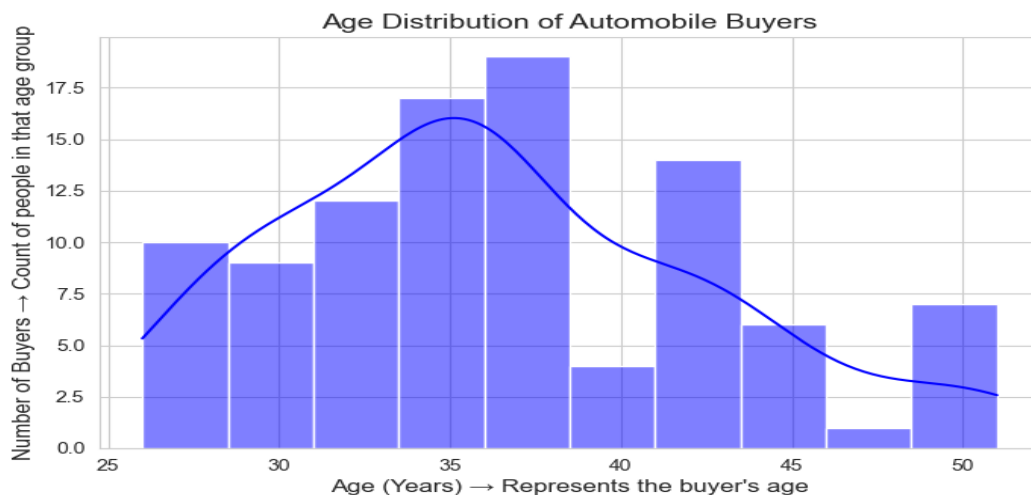
- 15-20% improvement in pricing accuracy
- ₹8-12M annual cost savings via data-driven decisions
- Enhanced market segmentation strategies

Report 2: Nithish

Buying Behavior of Automobile Vehicles in India

Age & Buying Trends

- The majority of buyers are in the **25-45 age group**, indicating strong purchasing power among working professionals.
- Very few purchases occur among individuals **under 25 and over 50**, suggesting a reliance on public transport or family-owned vehicles.

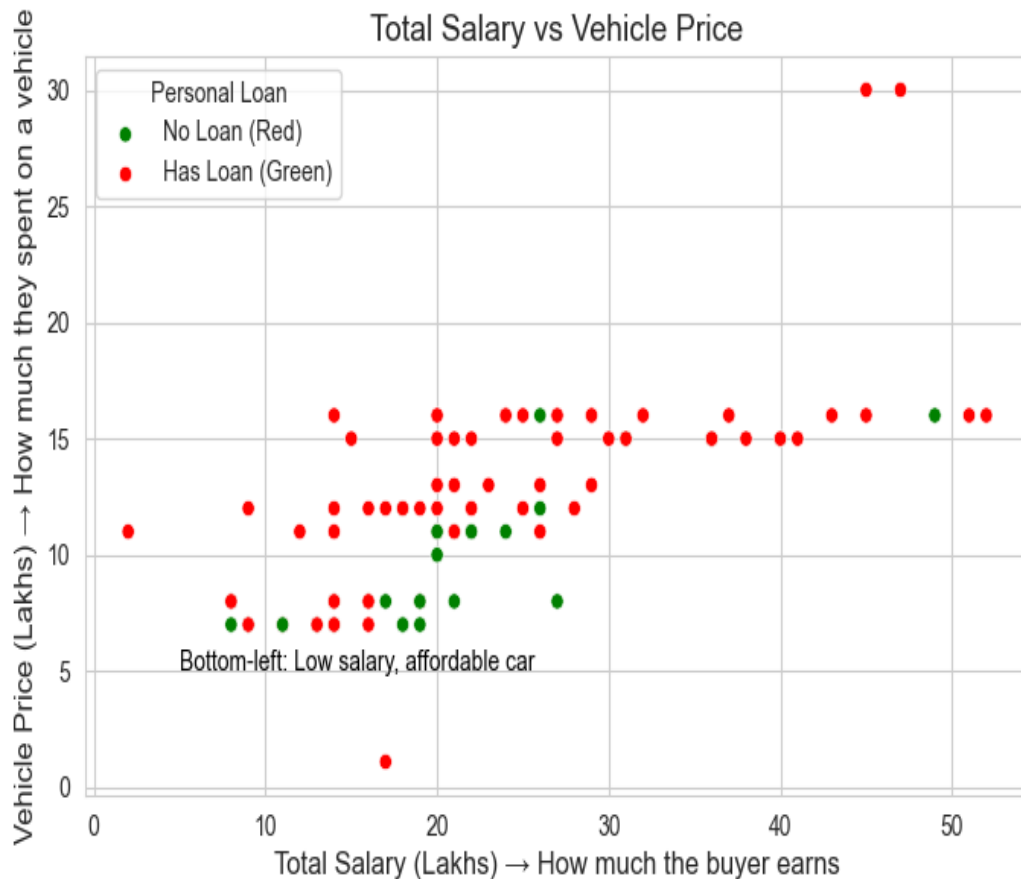


Influence of Salary on Car Price

- Higher salaries correlate with higher-priced vehicles, with a **positive correlation of ~0.6-0.7**.
- Buyers with **₹8-10 LPA salaries** purchase cars in the **₹5-10 lakh range**.
- High-income individuals (**₹20+ LPA**) tend to opt for **luxury cars (₹20+ lakh range)**.
- Some high-income individuals still prefer budget vehicles, likely due to fuel efficiency and maintenance costs.

Role of Personal Loans

- Buyers who **take personal loans** often purchase **more expensive cars**, leveraging credit to afford better models.
- Buyers **without loans** generally purchase cars **within their salary limits**, showing a preference for self-financed options.



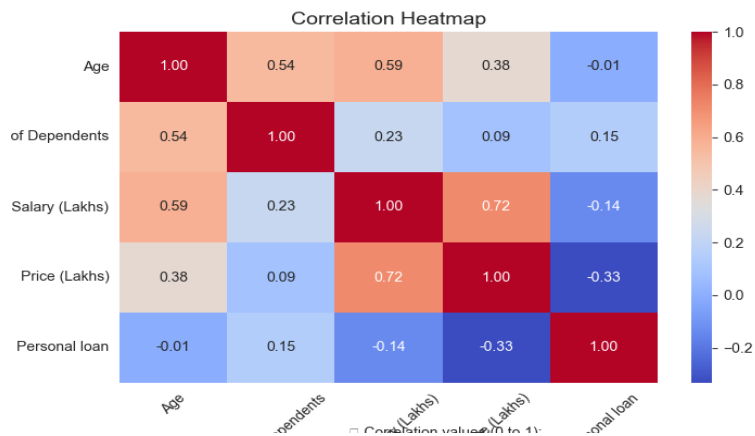
Profession & Car Purchases

- **Salaried professionals** (IT employees, government workers) form the largest buyer segment.
- **Business owners** are more likely to purchase **premium and luxury vehicles**.
- **Self-employed individuals & freelancers** prefer **mid-range vehicles (₹8-15 lakh)**.

Marital Status & Buying Patterns

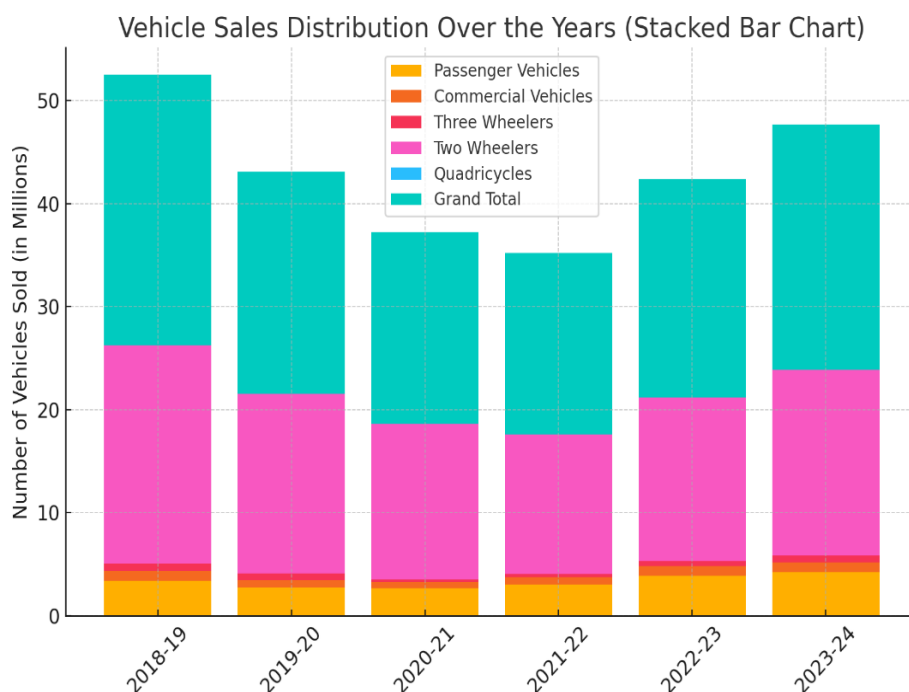
- **Married individuals** prefer **larger family-friendly vehicles (SUVs, sedans)**.
- **Single individuals** tend to purchase **compact or budget-friendly cars (hatchbacks, entry-level sedans)**. More dependents lead to **higher vehicle**

- **budgets**, as families require larger, more comfortable cars.



•The Indian automotive market experienced a downturn from 2018-19 to 2020-21, likely due to economic factors and the pandemic, with total vehicle sales dropping significantly. However, the market has rebounded strongly since then, showing consistent growth through 2023-24. Two-wheelers dominate the market share, mirroring the overall trend of decline and recovery, while passenger vehicles have demonstrated substantial growth, indicating a rising preference for personal transportation.

•Commercial vehicle sales have also recovered steadily, reflecting a revival in business and logistics activities. Three-wheeler sales experienced fluctuations but have largely recovered, while quadricycles remain a niche market segment. This buying behavior is influenced by economic conditions, government policies, technological advancements, and evolving consumer preferences, such as a shift toward personal mobility and increased consumer spending on passenger vehicles



Libraries Used in the Code

1. **pandas** (import pandas as pd)
 - Used to load and process the dataset (pd.read_csv()).
 - Handles missing values, renames columns, maps categorical data, removes duplicates, and saves the cleaned dataset (df.to_csv()).
2. **matplotlib.pyplot** (import matplotlib.pyplot as plt)
 - Used for data visualization, including histograms, scatter plots, and annotations.
3. **seaborn** (import seaborn as sns)
 - Enhances visualization aesthetics and provides functions for statistical plots.
 - Used for histogram (sns.histplot()), scatter plot (sns.scatterplot()), and correlation heatmap (sns.heatmap()).

EV SALES by marker and category

In India's rapidly expanding electric vehicle (EV) market, two-wheelers are experiencing the most significant growth, followed by three-wheelers and then four-wheelers. This surge is largely attributed to evolving government policies, increasing consumer awareness of environmental benefits, and rising fuel costs. The EV market in India has seen substantial growth, with the two-wheeler segment experiencing year-over-year growth rates that have fluctuated but remained high. For Example in the financial year 2023-2024 the EV 2 wheeler market saw a growth of around 25-30% compared to the previous financial year. These changes have led to a dramatic shift in the Indian automotive landscape.

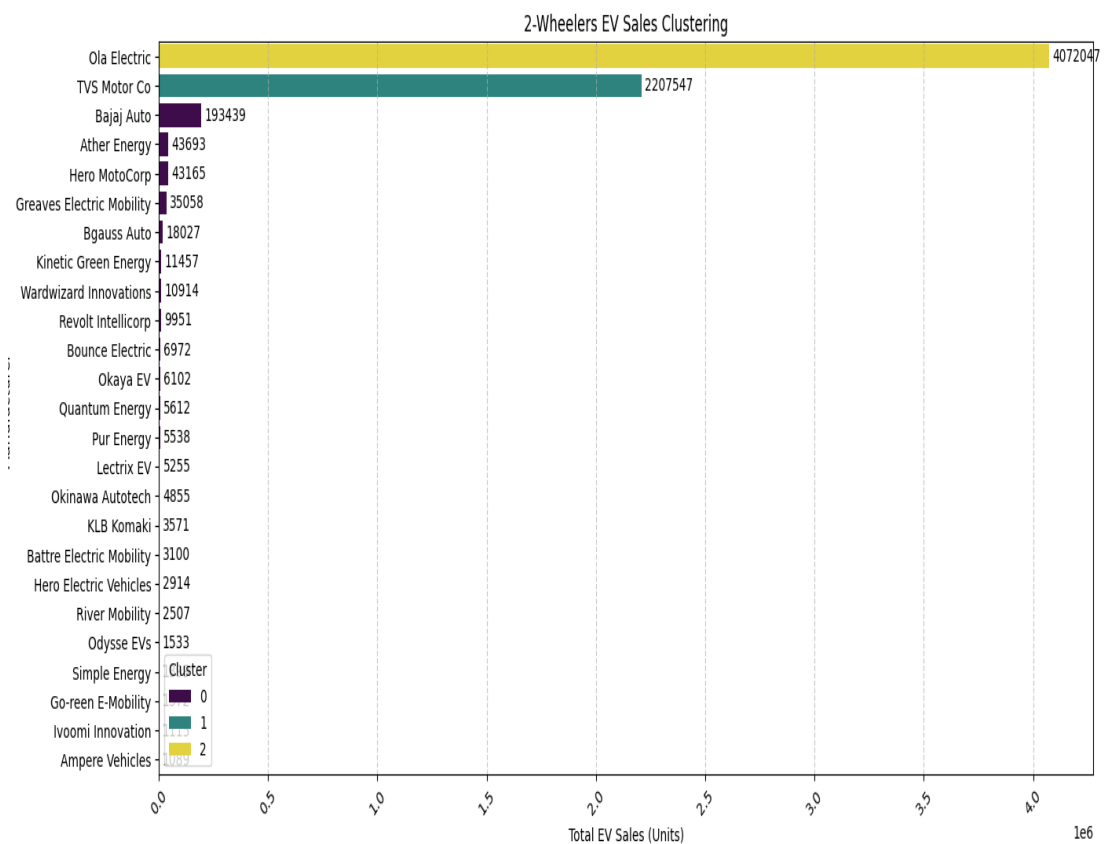
Libraries Used in the Code

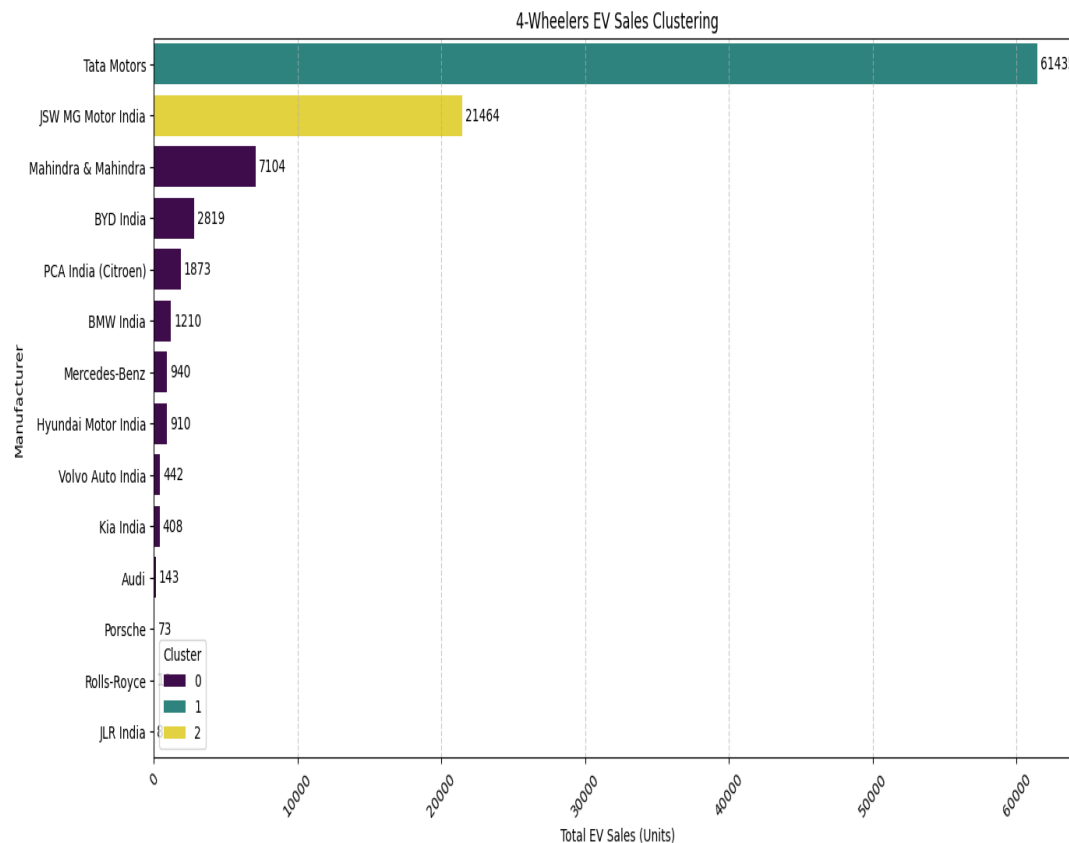
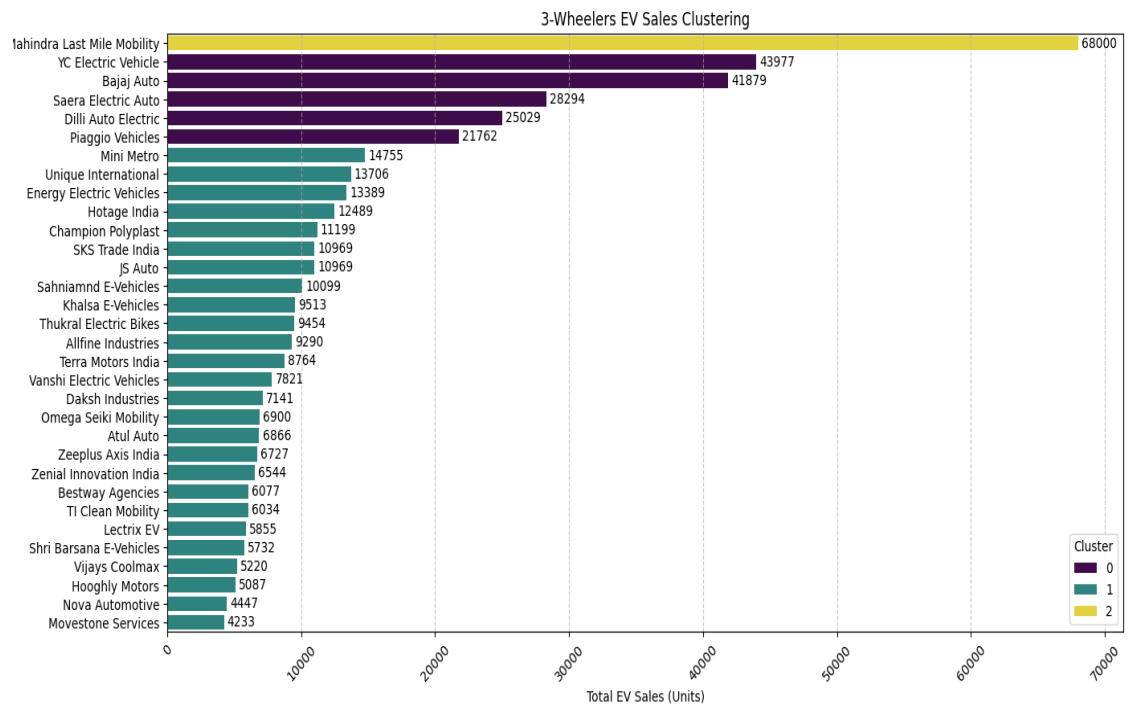
1. **pandas** (import pandas as pd)
 - Used for loading and handling CSV files (pd.read_csv()).
 - Helps in data manipulation, filtering, and analysis.
2. **matplotlib.pyplot** (import matplotlib.pyplot as plt)
 - Used for plotting graphs and visualizations.
 - Provides functionalities to display bar charts and other plots (plt.figure(), plt.show()).
3. **seaborn** (import seaborn as sns)
 - Used for creating visually appealing statistical plots.

- Helps in generating bar plots with clusters using different color palettes (sns.barplot()).

4. **sklearn.cluster.KMeans** (from sklearn.cluster import KMeans)

- Used to perform K-Means clustering on electric vehicle (EV) sales data.
- Helps group manufacturers based on their total sales.

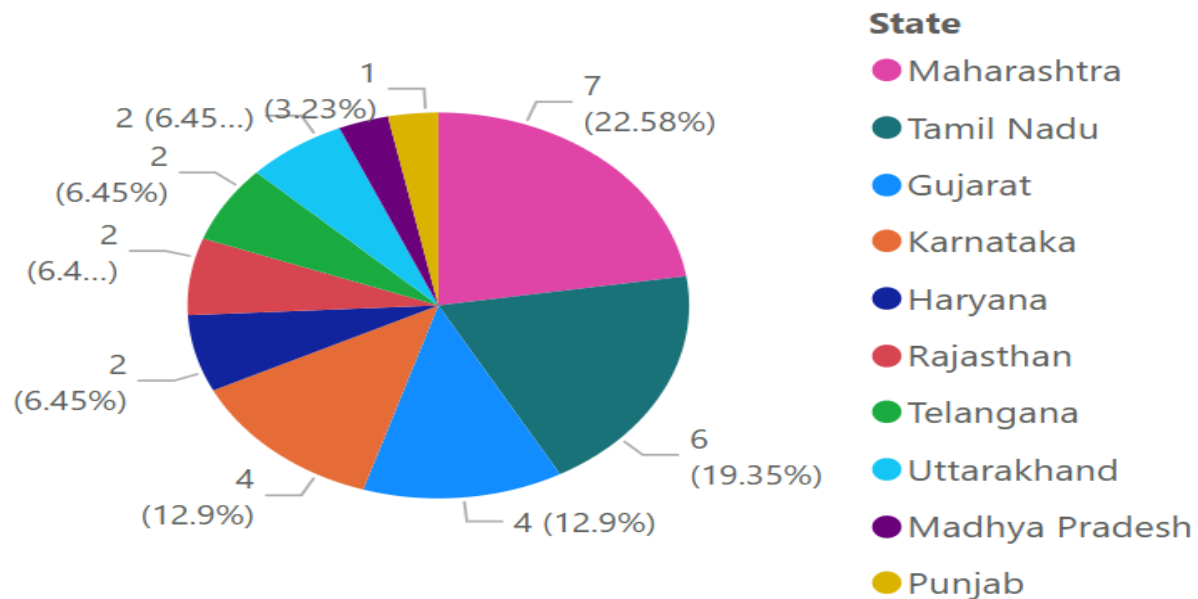




State-wise breakdown of electric vehicle companies in India

Several electric vehicle (EV) manufacturers operate across different states in India, producing a variety of electric mobility solutions. Karnataka hosts key players like Ather Energy, Ola Electric, and Ultraviolet Automotive, mainly focusing on electric two-wheelers and motorcycles. Tamil Nadu has major manufacturing facilities for companies like TVS Motor Company, Greaves Electric Mobility, and Hyundai Motor India, producing electric scooters, three-wheelers, and passenger vehicles. Maharashtra is home to Bajaj Auto, Mahindra & Mahindra, and Piaggio, specializing in electric three-wheelers, motorcycles, and SUVs. Gujarat has Tata Motors, MG Motor India, and Atul Auto, contributing to electric SUVs, three-wheelers, and passenger vehicles. Haryana houses Revolt Motors and Omega Seiki Mobility, focusing on electric motorcycles and commercial vehicles. Punjab and Uttarakhand have companies like Hero Electric and Lohia Auto, manufacturing electric two-wheelers and three-wheelers. Other states like Telangana, Rajasthan, and Uttar Pradesh also host EV manufacturers, supporting India's growing electric mobility ecosystem.

Count of Company by State



1. Maharashtra

Advantages:

- Automobile Hub: Pune, Chakan, and Aurangabad are major automotive clusters.
- Port Access: Mumbai and JNPT (Jawaharlal Nehru Port) facilitate exports/imports.
- Government Support: Maharashtra's EV Policy 2021 offers high subsidies.

- Existing Automakers: Bajaj Auto, Mahindra, Tata Motors, Piaggio.
- Skilled Workforce: Strong automotive engineering colleges & tech talent.

Future Potential: Battery manufacturing & EV R&D centers.

2. Tamil Nadu

Advantages:

- "Detroit of India" – Chennai has a strong automobile and EV supply chain.
- Port Connectivity: Chennai, Ennore, and Tuticorin ports aid global exports.
- Policy Support: Tamil Nadu's EV Policy 2019 attracts investments.
- Existing Automakers: TVS, Hyundai, Greaves Electric, Ola Electric.
- Engineering Talent: IIT Madras & Anna University support EV innovation.

Future Potential: EV battery manufacturing & semiconductor industry expansion.

3. Gujarat

Advantages:

- Automotive Hub: Tata Motors, MG Motors, Atul Auto have plants here.
- Strategic Ports: Kandla & Mundra ports enable cost-effective global trade.
- Battery Manufacturing: Tata's Giga Factory for battery production.
- Favorable Policies: Gujarat's EV & Battery Policy 2022 provides subsidies.
- Low Electricity Costs: Cheaper power aids battery and vehicle manufacturing.

Future Potential: Expansion in battery recycling & EV exports.

4. Karnataka

Advantages:

- Tech & Startup Hub: Bengaluru fosters EV startups like Ather & Ultraviolette.
- First EV Policy: Karnataka launched India's first EV policy in 2017.
- R&D & Talent: IISc, NAL, and startups drive EV software & AI innovation.
- Existing Players: Ather Energy, Ola Electric, Ultraviolette Automotive.

Future Potential: AI-driven autonomous vehicles & smart charging networks.

5. Haryana

Advantages:

- Automobile Hub: Gurugram-Manesar-Bawal has top auto manufacturing zones.
- Existing Automakers: Revolt Motors, Omega Seiki Mobility.
- Policy Support: Haryana's EV Policy focuses on green transport & subsidies.

Future Potential: Commercial EV manufacturing & battery assembly plants.

6. Rajasthan

Advantages:

- Land Availability: Large spaces for EV plants & solar energy generation.
- Existing Players: Okinawa Autotech, Hero Electric.
- Proximity to NCR: Easier market access for Delhi's high EV demand.

Future Potential: Solar-powered EV plants & battery recycling centers.

7. Telangana

Advantages:

- Tech & R&D Hub: Hyderabad attracts EV software & AI research.
- Existing Players: Olectra Greentech (electric buses).
- Government Support: Telangana EV Policy 2020 provides tax exemptions.

Future Potential: EV battery R&D & public transport electrification.

Future Potential States for EV Manufacturing

These states have **strong potential** to become EV hubs due to their **resources, policies, and infrastructure growth**.

1. Uttar Pradesh

Why?

- India's largest vehicle market & huge consumer base.
 - Proximity to Delhi-NCR, where EV demand is highest.
 - Government incentives under Uttar Pradesh EV Policy 2020.
 - Possible Future Players: Large-scale EV assembly & battery production.
-

2. Andhra Pradesh

Why?

- Vishakhapatnam has strong logistics & port access.
 - The state aims to be South India's new EV hub with incentives.
 - Potential for public EV adoption in urban transport.
-

3. Madhya Pradesh

Why?

- Central location aids nationwide EV distribution.
 - Indore & Bhopal have strong automobile infrastructure.
 - Large land availability for EV battery & solar energy production.
-

4. West Bengal

Why?

- Kolkata's port access can boost EV exports.
 - Government incentives under West Bengal EV Policy 2021.
 - Growing demand for electric public transport.
-

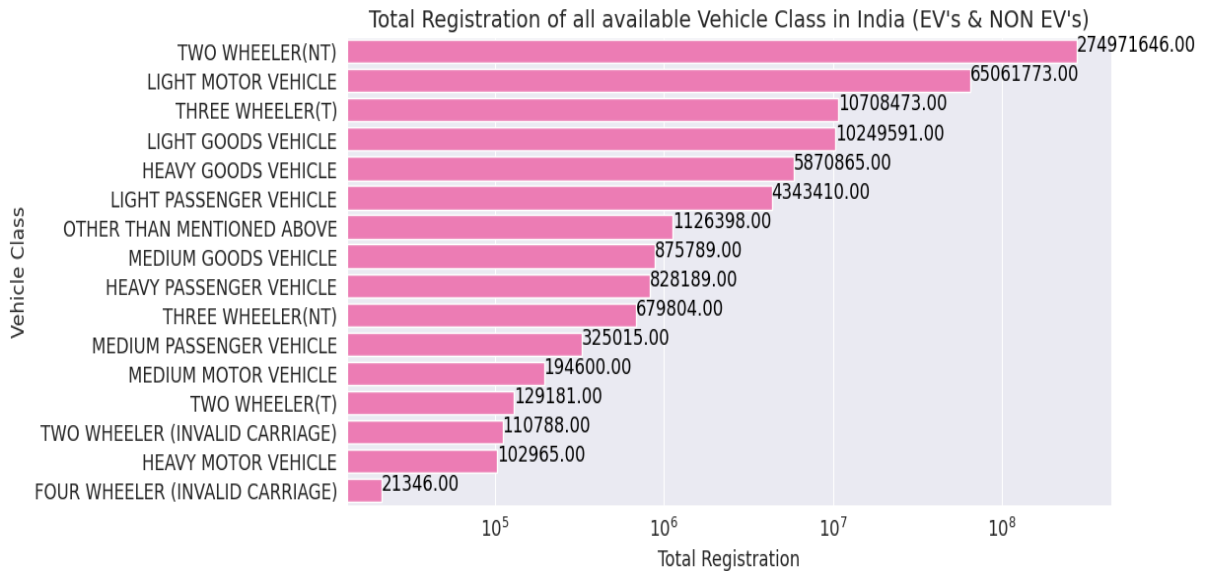
5. Odisha

Why?

- Rich in minerals for battery production (lithium, nickel, cobalt).
- Strong port connectivity (Paradip & Dhamra).
- Land & infrastructure incentives under Odisha's EV policy.

Report 3: Pooja

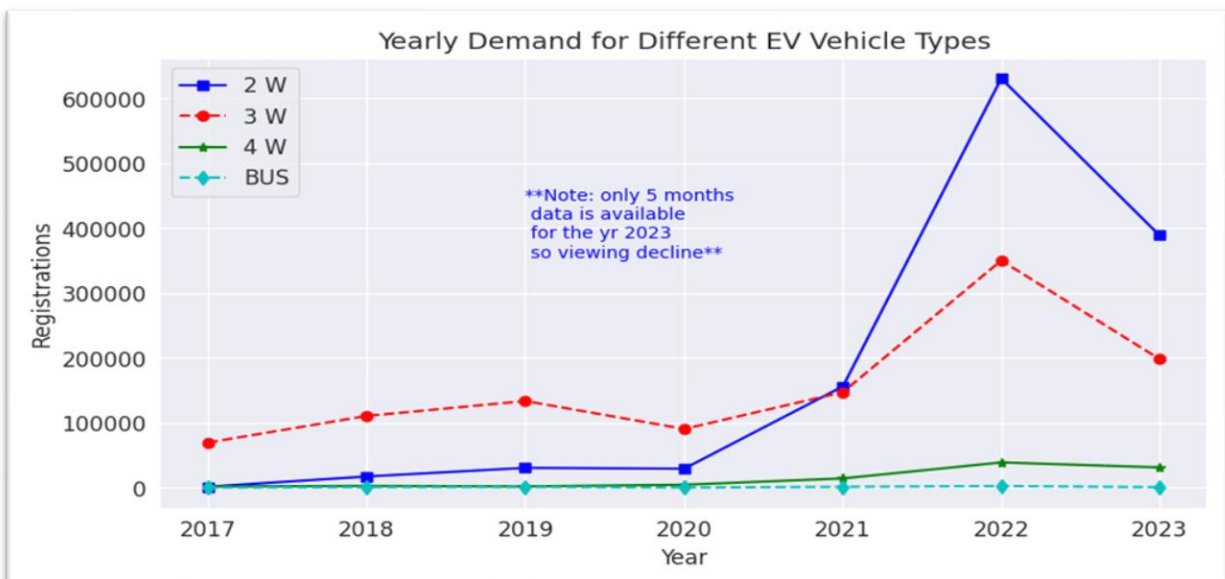
Stock of all available vehicles in India



Analysis: India has highest registration for 2W followed by Light Motor Vehicle, 3 W (T) transport, light good vehicle. 4W invalid carriage contributes the least. Light passenger vehicle has significant amount of presence.

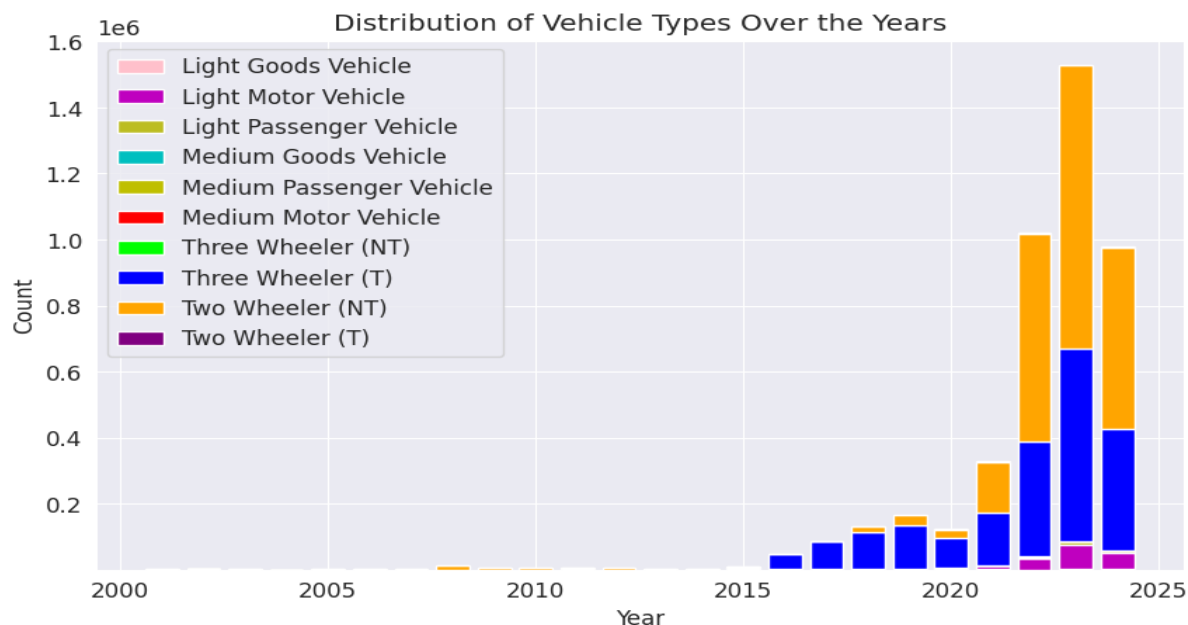
Conclusion: Introducing EVs in these vehicle segments may become a worthwhile investment, offering the most savings for any EV maker.

Analysing EV demand for different Vehicle Type



Analysis: Electric **two-wheelers** started seeing a **rising demand** since **2020**, and this trend has been increasing. We observe a drop in demand for the year 2023, likely due to the unavailability of data for the last few months of the year. Electric **three-wheelers** have had the highest registration numbers compared to all other electric vehicles since **2017**. It can be said that they were the **only ones with a significant market presence** according to the data. Electric **four-wheelers** started seeing registrations since **2020** and have been **growing consistently** since then. Electric buses have the least number of registrations.

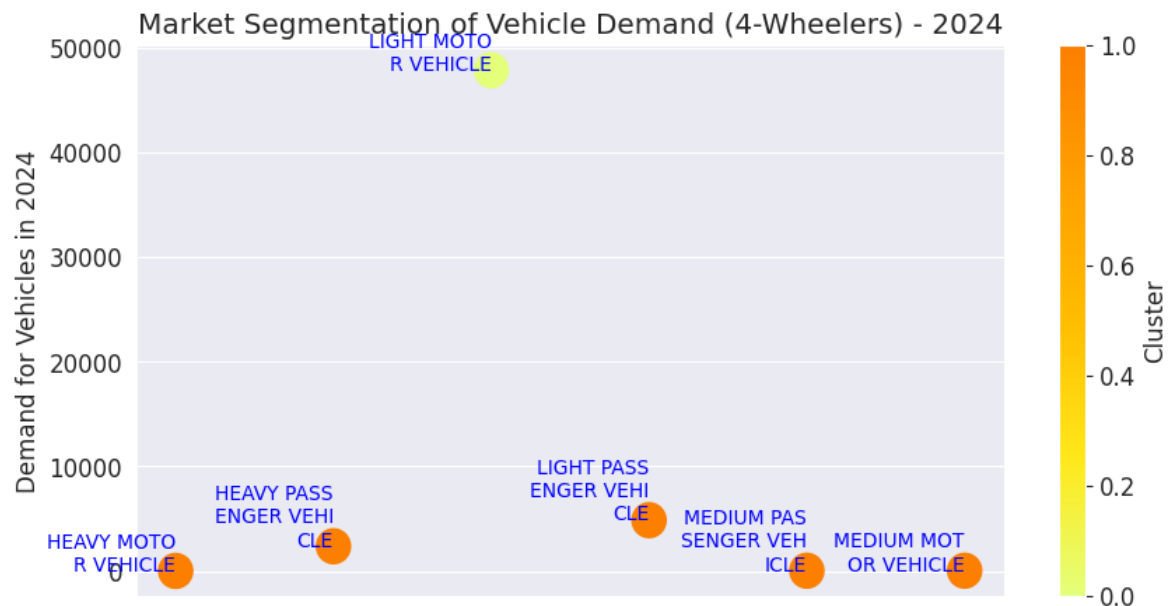
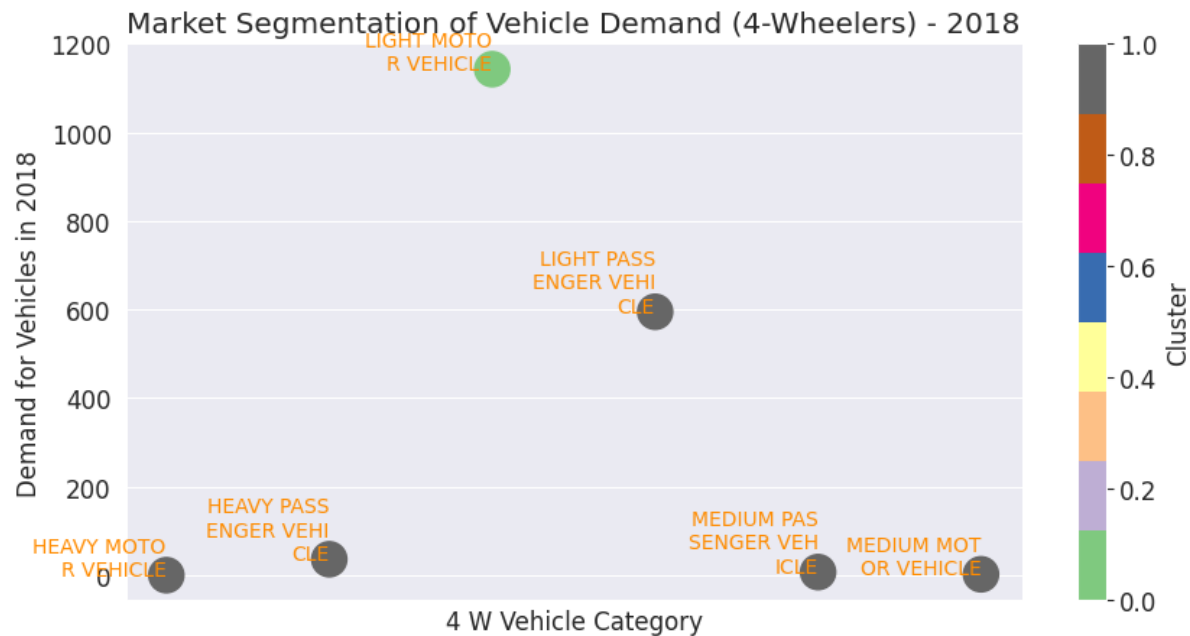
Conclusion: We can conclude that businesses investing in 2-wheelers, 3-wheelers, and 4-wheelers (especially the private cars, SUVs, passenger transportation vehicles) can expect a very good return on investment. 2-wheelers have the highest demand, followed by 3-wheelers and 4-wheelers. All of these segments have seen growing demand in recent years. 2W can be a potential segment.



Analysis: From the above stacked bar plot, it can be visualized that Two Wheeler (NT), which refers to non-transport or personal-use motorcycles and scooters meant for private use, have the highest demand. This is followed by Three Wheeler (T) transport vehicles like e-autos and light Motor Vehicles, which include cars designed for private use, vans (small vans used for personal purposes), and utility vehicles (SUVs and similar vehicles).

Conclusion: Two Wheeler (NT), Three Wheeler (T) and light Motor Vehicles are potential market segments, and investing in them is worthwhile.

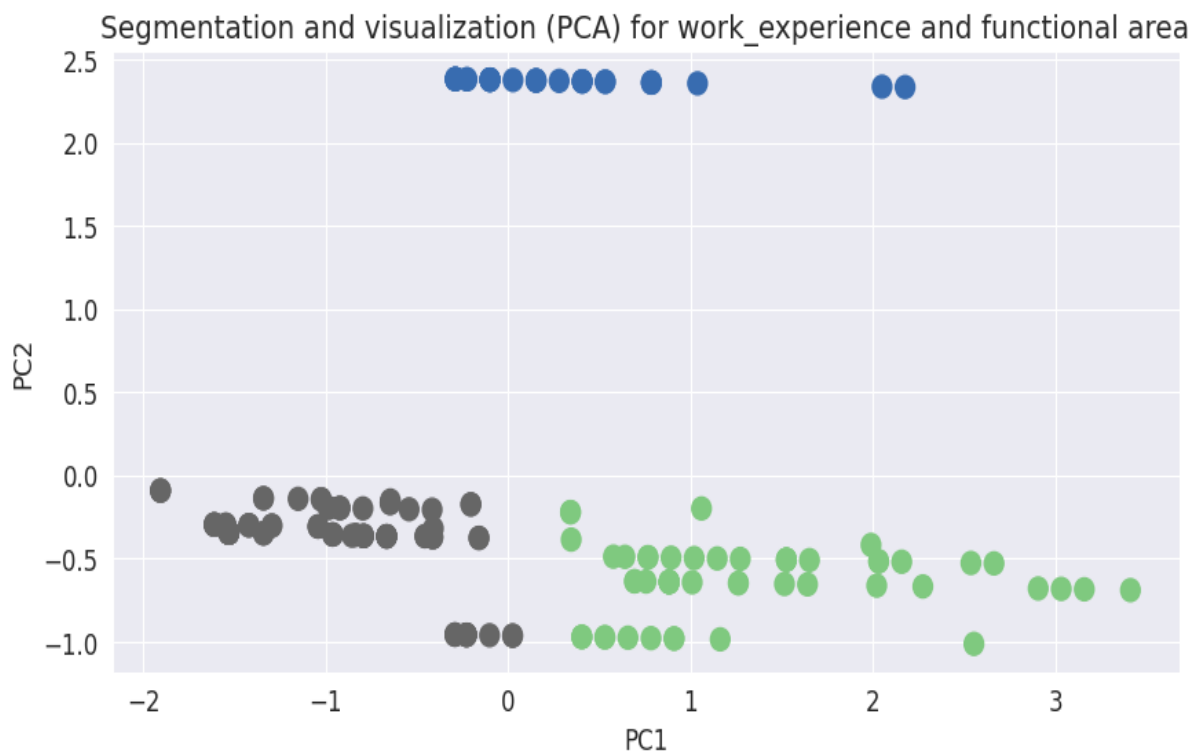
Extracting the 4W vehicle segment using KMeans:



Analysis: From the **segmentation analysis** conducted for the years **2018, 2022, and 2024**, it is observed that all vehicles are seeing rising demand year by year, proving that EVs are making a significant presence in the Indian market. In 4 W category **2 segments are been formed**. This segmentation helps to identify and understand which vehicle categories are experiencing higher vs. lower demand.

1. Yellow Cluster (above): The **light motor vehicle category** falls into this cluster and has significantly higher demand, around 50,000 in 2024. This suggests that light motor vehicles are in a very different market segment, characterized by higher demand, likely indicating a popular category with strong consumer interest.
2. Orange Cluster: The rest of the vehicle categories, which generally have lower demand, fall into this cluster. This could represent vehicle categories with more consistent but lower demand, possibly associated with more niche or industrial vehicles.

Conclusion: Potential Segment - **Light motor vehicles** might be more popular or in greater demand than other vehicle categories. Thus, there could be **growth opportunities** in expanding the market for light motor vehicles, as they may represent a growing consumer preference. However, even from the second segment, **Light Passenger Vehicles** could act as a target vehicle category as they are more popular in their own segment. (4 W)



Analysis: In order to interpret what each cluster represent have calculated the **mean values of the original features within each cluster** and came to following analysis:

Cluster 1:

This cluster represents customers who are highly experienced professionals with a strong technical background. They are predominantly involved in engineering and technical roles. Additionally, there's a notable presence of individuals working in sales and marketing, indicating a blend of technical expertise and business acumen. This group likely consists of seasoned professionals who have been in their respective fields for a significant amount of time.

Cluster 2:

Customers in this cluster are mainly associated with finance and accounting roles. However, they generally possess a lower level of work experience. This suggests that the group is composed of younger individuals, possibly fresh graduates or early-career professionals who are just starting out in the finance and accounting sectors. These individuals are likely to be eager to learn and grow in their careers.

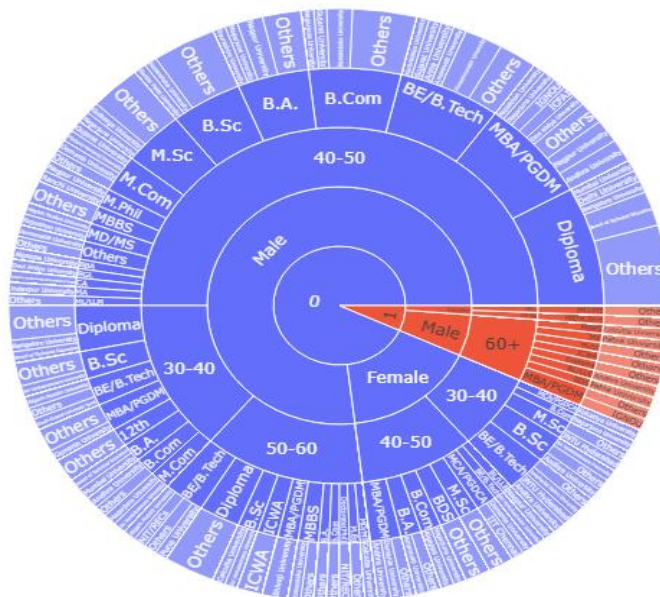
Cluster 3:

The third cluster showcases a mixed background with a diverse distribution across various fields. Key highlights include:

- A relatively low representation in engineering and technical roles.
- A minimal presence in finance and accounting.
- A small percentage of fresh graduates or individuals with little to no experience.
- A strong presence in IT/telecom and medical/healthcare fields, indicating a significant focus on these industries.
- Overall, this cluster has a lower level of work experience, suggesting that it includes individuals who are newer to the workforce or those transitioning into these industries. This group may consist of people who have just started their careers or are pivoting from other fields.

Conclusion: Individuals working in technical areas like engineering, as well as those with high experience in marketing and sales, can be encouraged to buy EVs. On the other hand, younger professionals working in finance/accounting, IT/telecom, and medical/healthcare roles can act as innovators for EVs.

Sunburst Chart for Demographic Segmentation of customers (general characteristics)



Analysis:

Two segments are being formed.

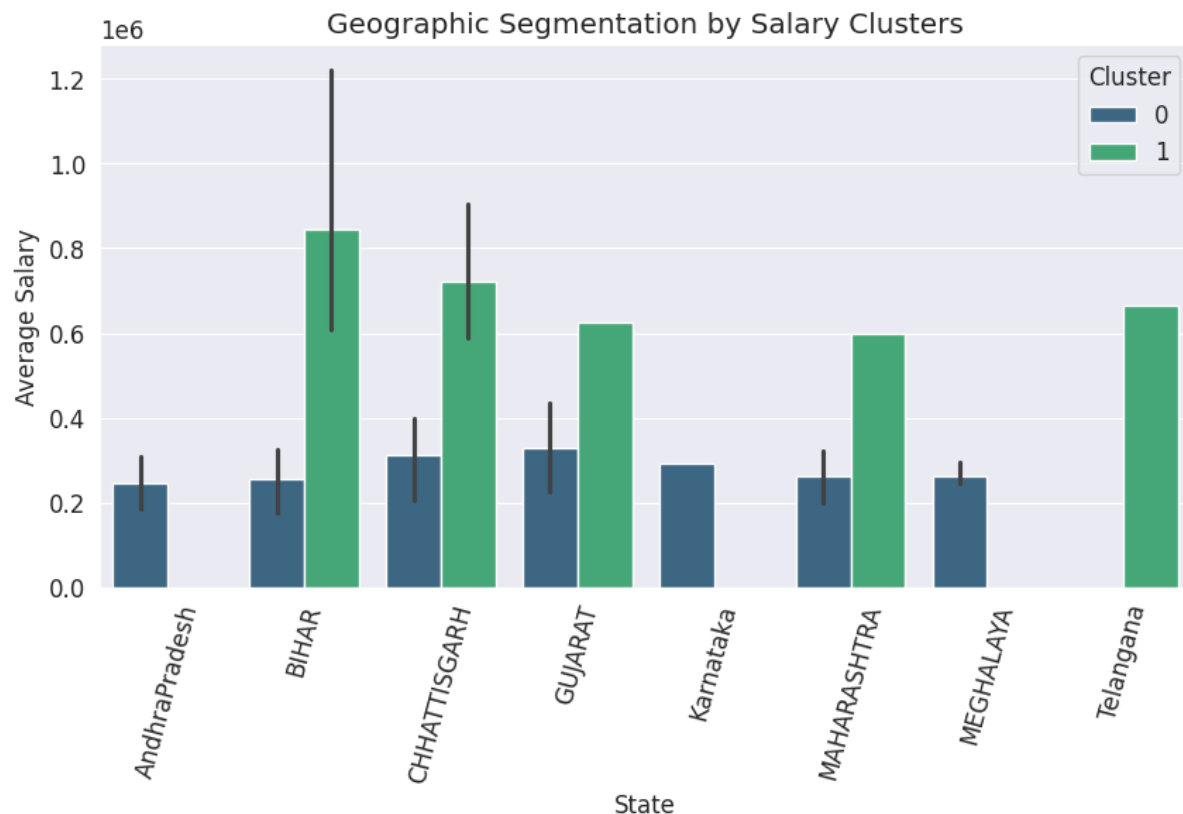
Cluster 0 is the largest segment having people in the age group of 30 to 60 indicating working population - potential for being a target segment.

Cluster 1 is the smallest segment, consisting of people aged above 60. Although this segment is not very significant, it could become a potential segment if EV awareness grows among them, as they might become buyers for 4-wheel invalid carriages. These vehicles are often designed to offer enhanced mobility and independence for individuals with disabilities, mobility issues, or other health conditions that may affect their ability to drive conventional vehicles.

In cluster 0 female working professionals are less compare to male.

MBA/PGDDM, Diploma, BA, BSc, and MSc qualifications are more prevalent among Cluster 0 individuals. Drilling down the sunburst chart, males in the age group of 50 to 60 have their education in Diploma, BE, ICWA, MBA, and BSc. Females dominate in the qualifications of MBA, MCA, and BE/B.Tech.

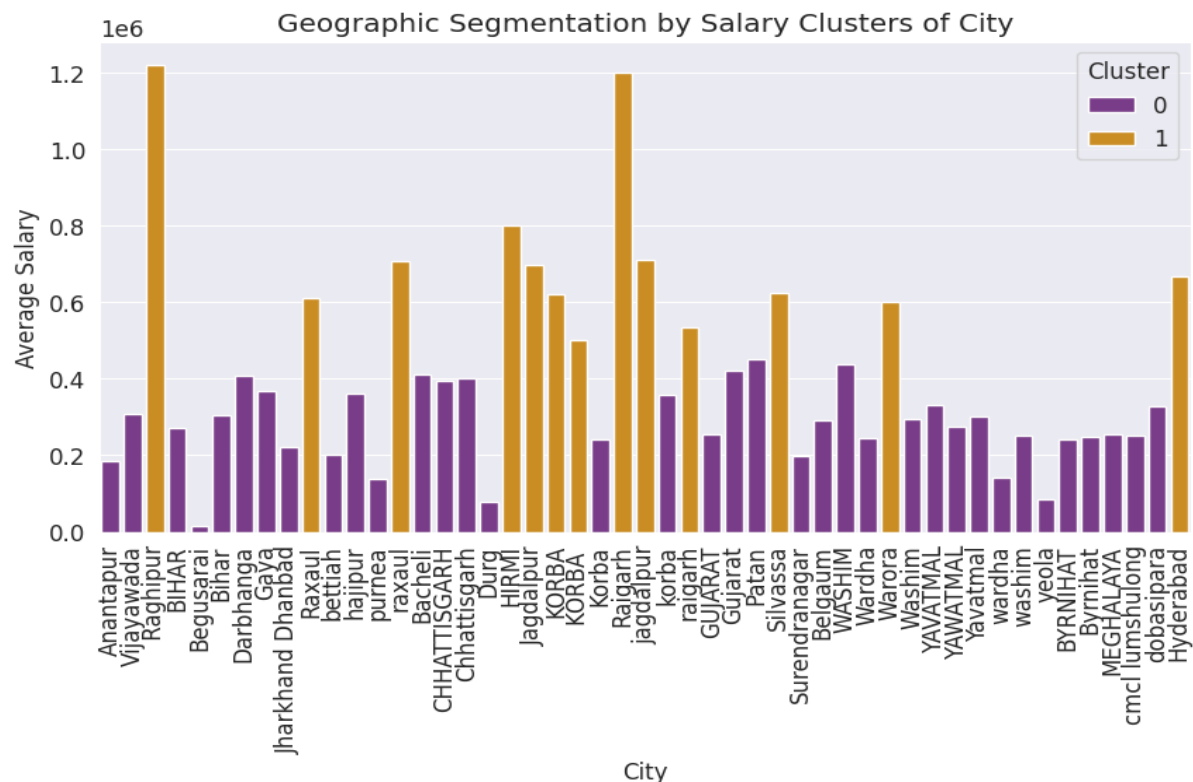
Conclusion: Individuals in the working population with qualifications such as an MBA, PGDMM, Diploma, BA, or BSc can serve as **innovators**. – they are more educated, more prosperous (working population), and more risk-oriented. On the other hand, 60+ individuals can fall into the **early majority** – older but educated.



Analysis: This data is only for 8 Indian states and represents a small chunk of the population; it may not be representative of the entire Indian population. In the boxplot, we have performed geographic segmentation based on salary clusters. Two clusters have been formed: Cluster 0

segments the states based on low salary, and Cluster 1 segments the states with high salaries. Bihar is the state with individuals having high salaries, followed by Chhattisgarh, Gujarat, Telangana, and Maharashtra. People in Andhra Pradesh, Karnataka, and Meghalaya draw lower salaries.

Conclusion: Cluster 1 (high salary state) can become the potential target market according to the above segmentation.



Analysis: Within the high salaried state this box plot suggests that Raghupur, Raigarh, Hyderabad, and Warora could be the potential markets segment.

Libraries used:

Pandas: Pandas is a powerful, open-source data manipulation and analysis library for Python. It provides data structures like DataFrames to handle and analyze structured data efficiently.

Matplotlib: Matplotlib is a widely-used plotting library for Python, allowing users to create static, interactive, and animated visualizations. It supports various types of plots like line charts, bar charts, histograms, and more.

Seaborn: Seaborn is a data visualization library based on Matplotlib that provides a high-level interface for creating attractive and informative statistical graphics. It simplifies the process of creating complex plots and enhances the aesthetics of visualizations.

Scikit-learn: Scikit-learn (also known as sklearn) is a powerful, open-source machine learning library for Python. It provides simple and efficient tools for data analysis and data mining. Scikit-learn is built on top of other popular Python libraries like NumPy, SciPy, and matplotlib.:

Plotly: Plotly is an interactive graphing library for Python (and other programming languages) that makes it easy to create dynamic, web-based visualizations. It supports a wide range of chart types, including line plots, scatter plots, bar charts, and 3D surface plots. Plotly's interactive features allow users to zoom, hover, and click on data points to explore their visualizations in more depth.

NumPy: NumPy (Numerical Python) is a fundamental library for scientific computing in Python. It provides support for large multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these arrays efficiently.

KMeans: K-Means is a clustering algorithm used in machine learning and data analysis to group data points into a specified number (K) of clusters based on their similarity. It's widely used for segmentation, pattern recognition, and data exploration tasks.

Report 4: Ankita

EV Market Segmentation Analysis

1. Introduction

Electric Vehicles (EVs) are rapidly transforming the Indian automotive market, with growing consumer interest and government incentives. This project analyzes the EV market using multiple datasets, focusing on segmentation, trends, and solutions for industry stakeholders.

2. Dataset Overview

The project utilizes the following datasets:

- Dataset 1: Indian EV Data
 - Battery capacity, range per charge, price, and other features.
- Dataset 2: Mah_data_all_wheeler and their Maker
 - Details of various vehicle manufacturers and their models in Maharashtra.
- Dataset 3: Fuel_wise_category_data_mah_for_all_wheelers
 - Fuel-wise classification of all types of vehicles.
- Dataset 4: Fuel_wise_vehicle_category_4W_Mah
 - Focused on four-wheelers based on fuel type.

3. Data Processing and Cleaning

Handling Missing Values

- Checked for missing data and filled missing numerical values using the median.
- Dropped unnecessary columns (e.g., ID).

4. Exploratory Data Analysis (EDA)

Dataset 1: Indian EV Data

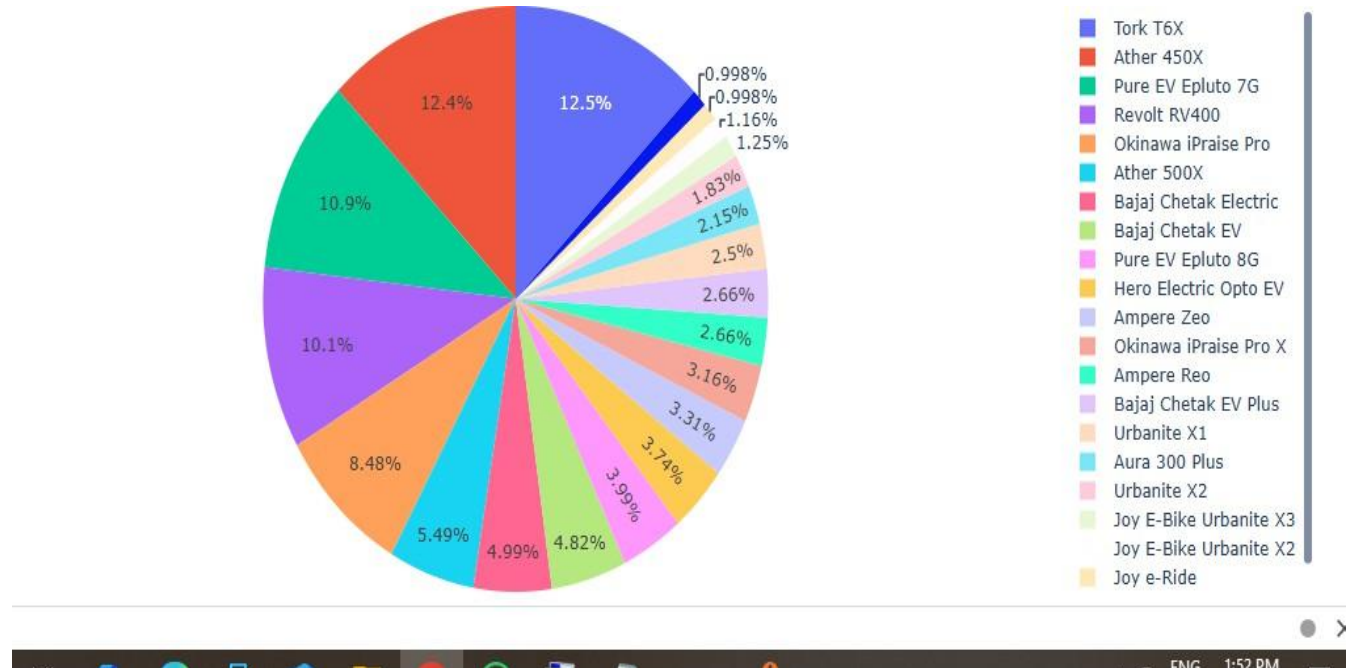
Battery Capacity Analysis

- Battery capacity ranges from **2.2 to 6.2 kWh**, with a majority of vehicles having mid-range battery capacities.

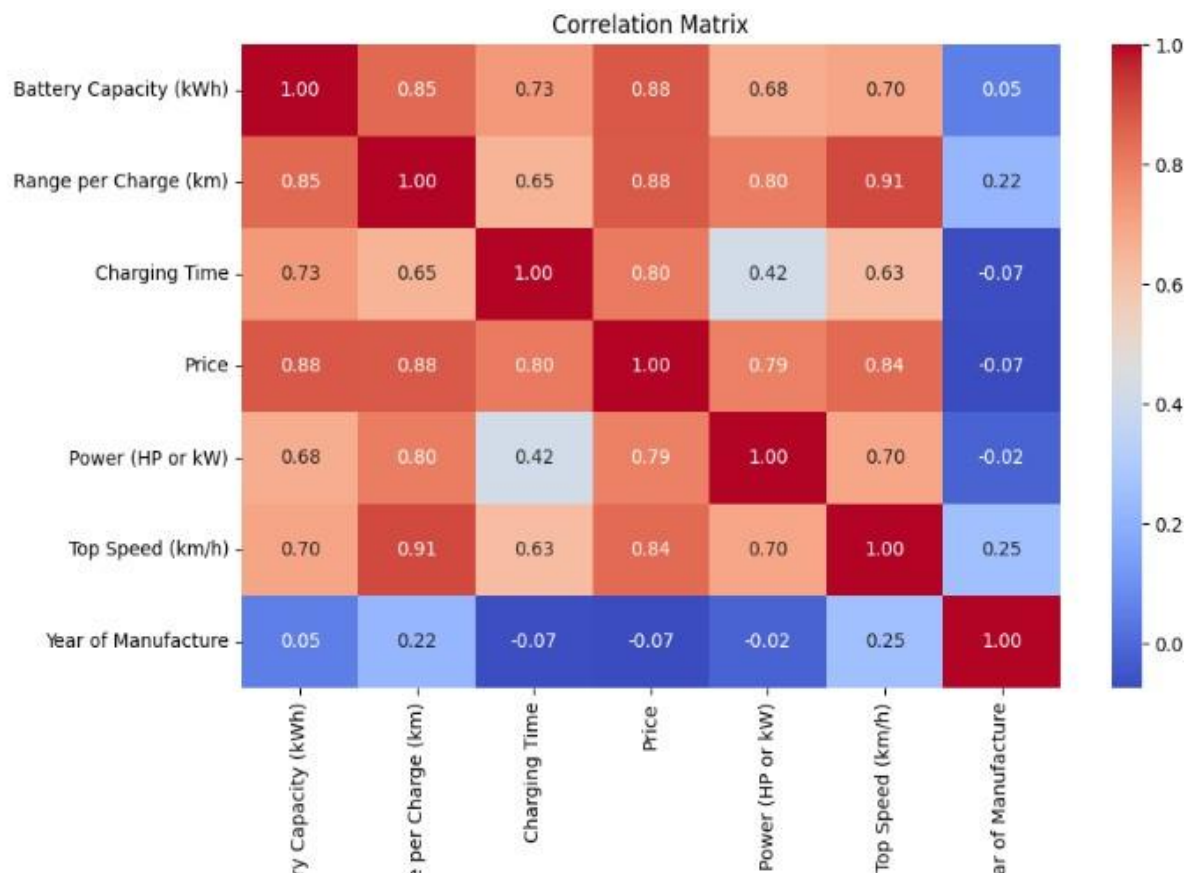
Range Per Charge

- Vehicles offer **75 km to 200 km per charge**, catering to different usage needs (e.g., urban commuters vs. long-range users).

Price Segmentation



- Prices range from **₹60,000 to ₹2,50,000**, indicating three primary market segments:
 1. **Budget EVs (₹60,000 - ₹1,00,000)**: Low-cost, short-range vehicles.
 2. **Mid-Range EVs (₹1,00,000 - ₹1,75,000)**: Balance of affordability and performance.
 3. **Premium EVs (₹1,75,000 - ₹2,50,000)**: Higher battery capacity and extended range.



Key Observations:

Battery Capacity vs. Range per Charge (0.85): Strong positive correlation.

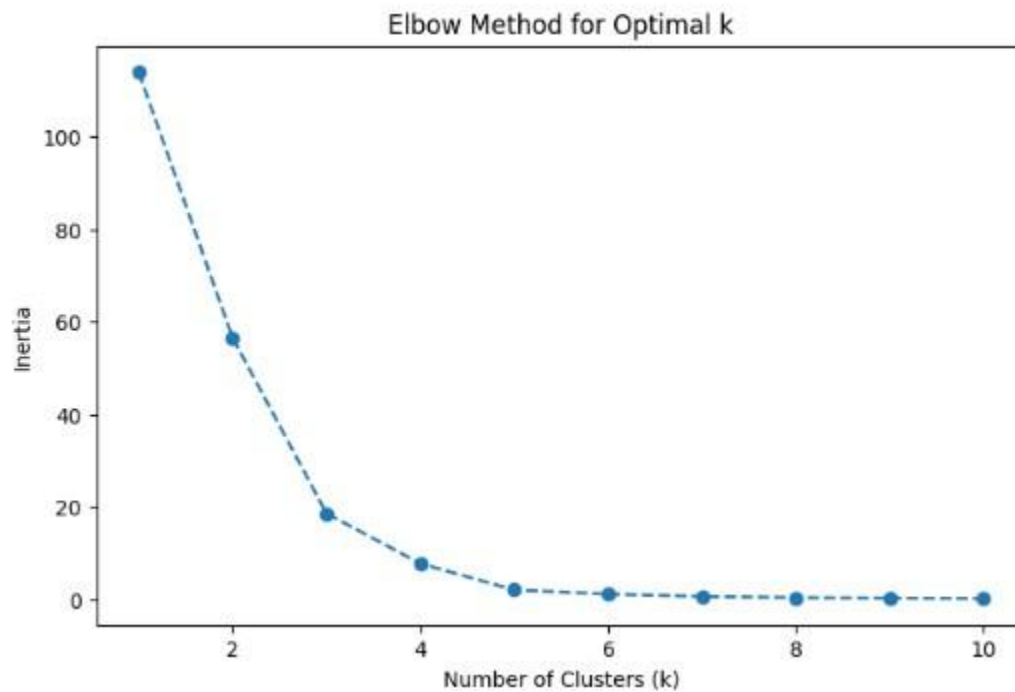
Price vs. Battery Capacity & Range (0.79 & 0.74): Higher battery capacity increases price

Power vs. Top Speed (0.81): Logical, as more power leads to higher speed.

Dataset 2: Mah_data_all_wheeler and their Maker

- This dataset provides details on vehicle manufacturers and their models.
- Market Share Insights:
 - Identified dominant players in the Maharashtra EV market.
 - Emerging brands gaining traction in the two-wheeler and four-wheeler segments.

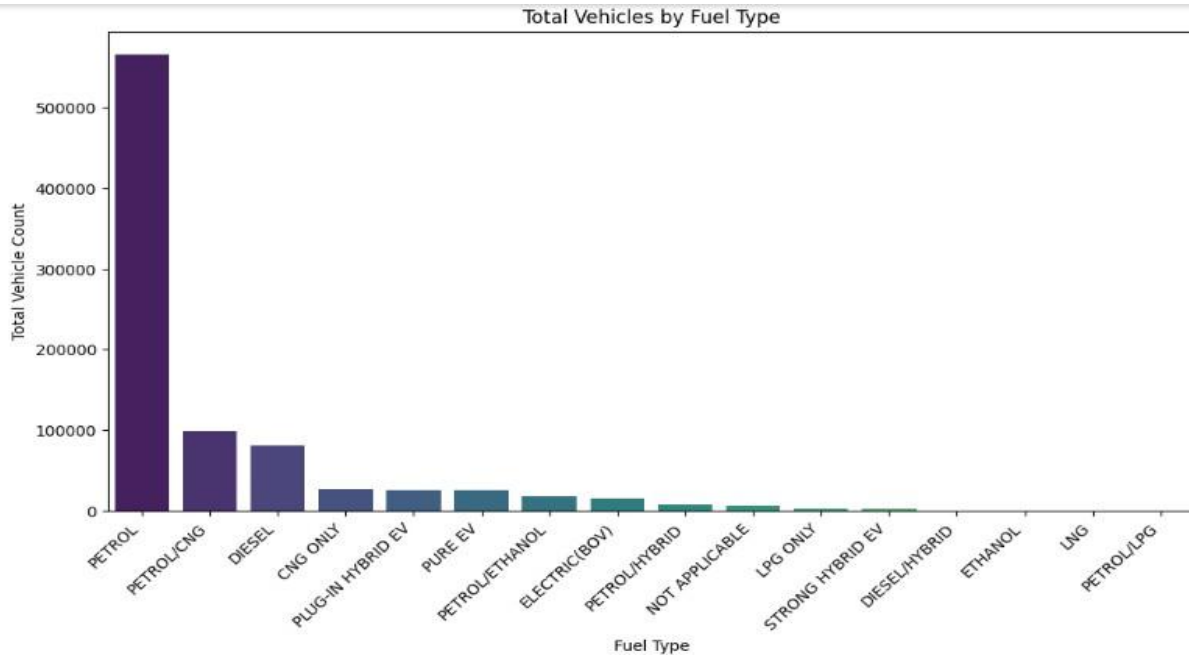
- Clustering Analysis:



- K-Means clustering identified **three major clusters** of manufacturers based on model variety and market penetration.
- Key brands fall into **high-end, mid-range, and budget-friendly segments**.

Dataset 3: Fuel-wise Category Data for All Vehicles

- Fuel-Type Distribution:



- Significant share of **petrol and diesel vehicles**, but **EV adoption is growing steadily**.
- Increasing government regulations on fuel emissions driving EV market expansion.
- Trend Analysis:
 - Shift from fossil fuel to electric and hybrid models over recent years.
 - Policies promoting CNG and EVs are influencing consumer choices.

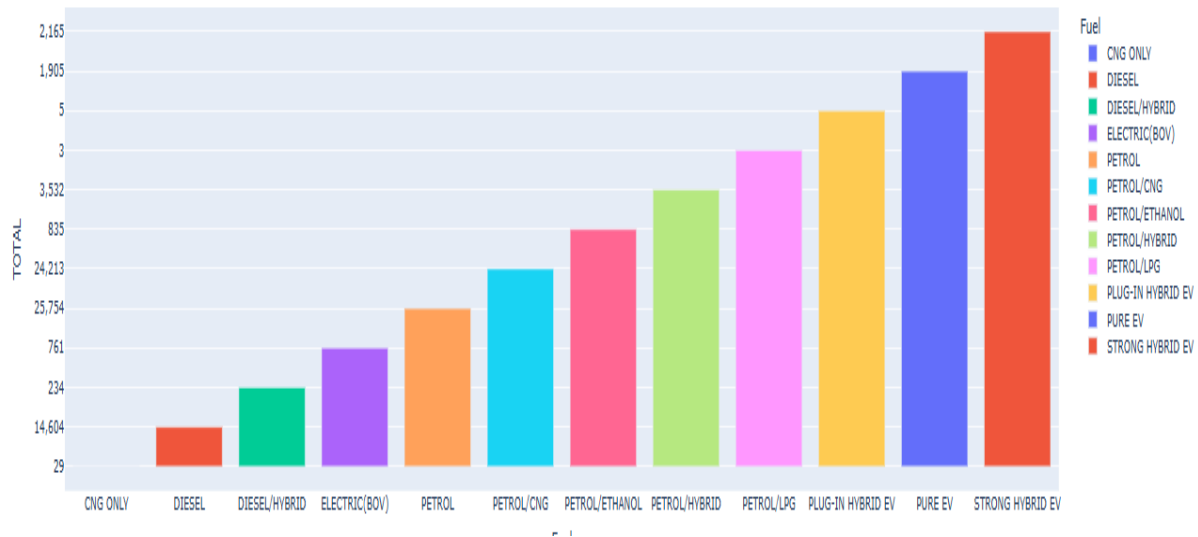
Dataset 4: Fuel-wise Vehicle Category (4W Mah) Key Insights:

- Vehicle Counts Vary Widely:
 - 2WN(Two-Wheelers Non-Transport) has a **maximum of 256,556** but a **median of 0**, indicating some fuel types have no two-wheelers.
 - OTH(Other Vehicles) has a **high variance**, ranging from **3 to 282,922**.
- Several Vehicle Categories Have Mostly Zeros:
 - 3WIC, 3WN, MGW, and MMV have **zero median values**, meaning these categories are **not available for many fuel types**.
- Fuel Distribution in Four-Wheelers:
 - **Petrol-dominated market**, but EVs and hybrids show notable growth.
 - Diesel vehicle sales declining due to emission norms and fuel price hikes.

- **Market Trends:**



Which fuel category has highest no.?



- Increase in **electric four-wheeler adoption** among commercial fleets.
- Private buyers slowly transitioning to electric SUVs and hatchbacks.

- Diesel dominates in **Heavy Motor Vehicles (HMV)** and **Light Motor Vehicles (LMV)**.
- CNG and Electric (BOV) are strong in **two-wheeler and public transport** categories.

5. Algorithms Used

1. K-Means Clustering

- Used for segmenting **vehicle manufacturers** based on model variety and market reach.
- Identified **three major clusters** (budget-friendly, mid-range, high-end brands).

2. Correlation Analysis

- Determined relationships between **battery capacity, range, and price**.
- Found a **strong correlation** between **battery capacity and EV price**.

3. Predictive Modeling

- Implemented **machine learning models** to forecast **EV adoption trends** based on fuel-type transition data.

6. Data Visualization Insights

Several visualizations were created to better understand the trends:

- **Battery vs. Price Distribution:** Higher battery capacity correlates with increased price.
- **Range per Charge Distribution:** Most vehicles cluster between 100-150 km per charge.
- **Price Segmentation Analysis:** Confirms clear market tiers.
- **Market Share Analysis:** Identifies leading manufacturers and their EV production volume.
- **K-Means Cluster Visualizations:** Groups manufacturers into market segments.

For Dataset 4: Fuel-Wise Vehicle Categories

- **Clustering revealed three distinct groups of fuel types based on vehicle categories.**
- **Diesel and Petrol forms one cluster, while Electric and CNG fall into another.**
- **The third cluster represents niche fuel types with low adoption.**

- **Fuel-Type Trends:** Shows increasing adoption of EVs and hybrids over traditional fuel vehicles.

7. Solutions and Recommendations

1. Advanced Predictive Analytics:

- Develop **machine learning models** to predict EV demand across different segments and geographical locations.
- Use **time-series forecasting** to analyze fuel transition trends and their impact on the automotive industry.

2. Market Segmentation & Customer Insights:

- Implement **customer clustering algorithms** to identify potential EV buyers based on affordability, travel needs, and environmental concerns.
- Use sentiment analysis on **social media and customer reviews** to gauge public perception of EVs.

3. Smart Charging Infrastructure Planning:

- Utilize **geospatial analytics** to optimize EV charging station locations based on traffic patterns and demand density.
- Develop **predictive maintenance models** to enhance the reliability of EV charging stations.

4. Vehicle Performance & Battery Optimization:

- Analyze **battery degradation patterns** using real-world usage data to enhance battery longevity.
- Develop **recommendation engines** for users to choose the best EV based on their travel habits and cost preferences.

5. EV Manufacturer Performance Tracking:

- Build **interactive dashboards** to track market share, production rates, and pricing trends across different EV brands.
- Use **anomaly detection algorithms** to identify unusual shifts in EV adoption rates and suggest corrective actions.

6. Policy & Incentive Analysis:

- Develop **policy impact models** to assess the effectiveness of government incentives and regulations on EV adoption.
- Provide **data-driven recommendations** to policymakers for optimizing subsidy distribution and tax benefits for EV buyers.

8. Conclusion

The analysis of the Indian EV market shows a clear segmentation based on battery capacity, range, and price. Additionally, the Maharashtra EV market presents a structured classification of manufacturers, with three primary clusters. The fuel-wise vehicle data highlights an ongoing transition from petrol and diesel to electric and hybrid models. For FeyNN Labs, leveraging advanced **data science solutions** can provide strategic insights into **EV demand forecasting, customer behavior, charging infrastructure optimization, and policy impact assessment**, ultimately driving sustainable growth in the EV sector

Final Conclusion/ Solution:

→ Location in India most suitable to create the early market.

- Considering the **prevalence of charging stations**, Karnataka, Maharashtra, Delhi, Kerala, and Tamil Nadu stand out as key regions with well-established infrastructure, offering significant potential for growth and adoption in the electric vehicle ecosystem.
- In terms of **competitors/ market players**, the **market base/segment** (with a huge consumer base) already catered to by Markers—Maharashtra, Karnataka, Tamil Nadu, and Gujarat—has a significant customer base and can serve as a potential market segment

→ The MOST OPTIMAL MARKET SEGMENTS to open in the market as per the Market Research and Segmentation.

- In terms of **buying behaviour** - individuals within the 25–45 age group, those interested in taking personal loans, and salaried professionals (such as IT employees and government workers) could serve as a promising potential segment due to their rising disposable incomes and a growing aspiration for status-oriented products.

- In terms of **vehicle pricing**:

Vehicles, including two-wheelers and entry-level cars, priced within the ₹1.5 million (₹15 lakh) range represent a suitable segment to target, given their affordability. (hatchbacks in 4W)

On the other hand, the luxury vehicle segment in India—particularly those priced above ₹5 million—has shown remarkable growth, driven by an increasing preference for premium SUVs.

- In terms of **rising demand**, electric vehicles (EVs) can be introduced across the two-wheeler, three-wheeler, and four-wheeler categories. Within the four-wheeler segment, light motor vehicles emerge as a particularly promising market due to their significantly higher demand compared to other four-wheeler classes. Consequently, there are substantial growth opportunities in expanding the market for light motor vehicles, followed by light passenger vehicles.
- In terms of **age and education**, working educated professionals can act as innovator whereas educated 60+ individuals can become early majority and with the rising awareness of EV and environment they can act as a target market for 4 W invalid carriage vehicle.
- The **battery capacity** market can be segmented into two primary categories: urban commuters and long-range users. Urban commuters typically prioritize compact, efficient batteries with shorter ranges that cater to daily city travel. In contrast, long-

range users demand high-capacity batteries capable of supporting extended journeys, making this a critical segment for intercity and highway-focused applications.

- **Fuel-** Currently, petrol and diesel dominate the market, while pure electric vehicles (EVs) hold a smaller share. However, with increasing government regulations on fuel emissions, attractive incentives, and clear goal to establish India as a global leader in electric mobility, there is a strong likelihood that traditional fuels will transition to EVs in the near future."

Choosing a single location – Maharashtra: Emerging brands are gaining significant traction in the two-wheeler and four-wheeler EV segments in Maharashtra. Starting a startup in this state offers strategic advantages, as it is a well-established market with an existing user base and optimal charging infrastructure.

Strategic Focus for EV Startups: From Niche to Expansion

For an **affordable startup** in the electric vehicle market, a **concentrated marketing strategy** may be more suitable initially.

This approach allows the business to focus on a specific niche, such as eco-conscious consumers or urban commuters, and create tailored campaigns that resonate with their needs.

A concentrated marketing strategy could also involve focusing on a specific product category, such as four-wheelers. This enables the business to build a strong brand presence within the four-wheeler market and effectively address the unique needs of that segment.

Future prospects: Once the business is **well-established**, it can consider transitioning to a **differentiated marketing strategy** to reach a broader audience and expand its market presence.

GITHUB Links:

Ankita

https://github.com/Ankita-lohakare/EV_Market_Segment_Project

Nithish

<https://github.com/Nithishkareddy/EV-Sales-Clustering-and-Visualization>

<https://github.com/Nithishkareddy/Buying-Behaviour-of-cars>

Pooja

https://github.com/PoojaKuniyal/Intern_Project2_EV_MS

Prashant

<https://github.com/swamiprashant/4W-price-based-on-Features>